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MASTER IN SPACE ENGINEERING



3D Design and Study of Lunar Landing Pads using CFD Analysis Software

Master Thesis

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Abstract—As humanity extends its reach towards extraterrestrial exploration, the necessity of establishing safe and efficient landing infrastructure becomes paramount. This master's thesis delves into the intricate domain of lunar landing pad design, addressing the unique challenges posed by the lunar environment. The study employs computational simulations and analyses to scrutinize various landing pad configurations. Results from the simulations are presented and discussed, highlighting the performance variations among the analyzed landing pad designs, aiming to provide valuable insights to the emerging field of lunar infrastructure development. This master's thesis represents a contribution to the field, offering a further point of view of the intricacies involved in lunar landing pad design.

I. INTRODUCTION AND OBJECTIVES

Understanding the world around us is the biggest challenge the human being has ever faced. In order to reach this objective, the development of different tools has characterized the civilization throughout history. The technology improvement has always been tightly related to the evolution of human reasoning and intelligence. Despite the development of technology had the aim of enhance the survival capabilities of humans, nowadays, it represents, among other things, the revolution of the cognitive faculties necessary to fulfill one of the most important human traits - curiosity.

The existence of planets has intrigued humans since the early stages of civilization. Outer space and its uncertainties have played a crucial role in the development of astronomy, mathematics, astrophysics, and engineering. To explore beyond Earth's atmosphere, scientists have confronted challenges spanning various fields of study. In recent decades, these scientific fields have experienced remarkable growth, thanks to technological improvements, especially driven by the United States (NASA) and the USSR/Russia (Roscosmos) in the 20th century.

The Cold War tensions between the United States and the Soviet Union in the second half of the 20th century led to a rapid development in defense and military power, ultimately sparking the space race. In 1957, Roscosmos made the first move by launching Sputnik, the first artificial satellite orbiting Earth. In 1961, the Soviets achieved another milestone by sending the first human, Yuri Gagarin, into space, completing one orbit around the Earth in a historic 108-minute flight [1].

NASA caught up in the space race during the 1960s, leading to the development of Project Apollo, which successfully landed astronauts on the lunar surface between 1968 and 1972. Since then, no astronaut has set foot on the lunar surface again, however, the Moon kept being a matter of increasing interest for the different space agencies. Despite any crewed lunar mission has been carried out, numerous non-crewed missions such as BioSentinel from NASA, Chang'e from CNSA, and Chandrayaan from ISRO (the first to land near the lunar South Pole [2]), have contributed to lunar exploration.

The lunar surface is a hazardous environment that needs to be studied carefully prior to planning any exploration mission. The lack of atmosphere determines the Moon surface to be under high vacuum conditions, becoming a very dangerous environment, not only because humans are not able to survive under low pressure conditions, but also because the lack of protection from the cosmic and Sun radiations does not prevent these radiations from reaching the lunar surface. The radiation that incises on the Moon heats up considerably the surface region that is facing the Sun, in contrast, the absence of an atmosphere makes the accumulated heat to escape from the lunar surface when the region is not illuminated, hence decreasing the temperature drastically. This generates enormous temperature fluctuations, like the ones measured at the Apollo 15 site where temperature oscillate between 100°C and -180°C approximately [3]. Despite the environmental conditions of the Moon represent some of the main constraints when planning a mission, they are not the most worrying factors.

The lunar dust has posed several significant challenges for space exploration missions, especially during manned missions to the Moon. Some of the issues associated with lunar dust include abrasion and mechanical damage, contamination, sealing and leakage issues, interference with instruments and equipment, adherence that induces human health danger, etc. As remarked by Gene Cernan in the Apollo 17 Technical Debriefing [4]:

"I think dust is probably one of our greatest inhibitors to a nominal operation on the Moon. I think we can overcome other physiological or physical or mechanical problems except dust and its adherence to everything no matter what kind of material... Once you get inside the spacecraft, as much as you dust yourself, you start taking off the suits and you have dust on your hands and

your face and you're walking in it. You can be as careful in cleaning up as you want to, but it just sort of inhabits every nook and cranny in the spacecraft and every pore in your skin."

Efforts to address the lunar dust challenge have become a focal point for future lunar exploration missions. Researchers and engineers are actively exploring innovative solutions, such as advanced sealing technologies, improved cleaning procedures, and the development of specialized materials resistant to lunar dust adherence.

One of the most popular paths of research is the concept of Lunar Landing Pads (LLP), surfaces designed to mitigate the impact of lunar dust during spacecraft landings and takeoffs. LLP represent a crucial step in addressing the unique conditions of the Moon's surface. Several space agencies, including NASA and international partners, are actively investing in research and development to create effective Lunar Landing Pads. These initiatives involve the research in different fields such as materials science, aerospace engineering and planetary geology to design surfaces that can withstand the harsh lunar environment.

As humanity prepares for a new era of lunar exploration through programs like Artemis, the request for effective solutions to lunar dust-related issues exemplifies our ongoing commitment to pushing the boundaries of space exploration and expanding our understanding of celestial bodies beyond Earth.

In accordance to what has been said, the purpose of this project is to propose a design for a Lunar Landing Pad, taking inspiration from previous designs explored by other authors. It will address the problematic of lunar dust during rocket landings and take offs, and, by means of computational fluid dynamics (CFD) analysis, optimize the shape in order to improve the performance and reduce the material required. Therefore, reducing the amount of time and resources needed to build the LLP.

This document will have the following structure: The state-of-the-art section, where the main characteristics of the lunar environment, lunar regolith and plume-surface interaction will be exposed as well as a review of many previous LLP designs and other mitigation ideas. The methodology section, that will describe the process of the design and its iterations, together with the procedure of the CFD analysis, determination of boundary conditions, material

data and load cases. The results and discussion section, where the results obtained by the analysis will be exposed and discussed, and, finally, the conclusions of the project and the future work.

II. STATE-OF-THE-ART

The challenge posed by the exhaust gases of a rocket during lunar landings or take-offs is highly relevant, primarily due to the acceleration of lunar surface particles such as dust, sand, gravel, and rocks to extremely high velocities. The magnitude of these velocities depends on both the size of the affected particles and the speed of the exhaust gases in the plume. Furthermore, the absence of an atmosphere on the Moon exacerbates this challenge.

In the case of powerful rocket engines, the regolith on the Moon's surface can be accelerated to lunar orbital velocities. This results in the formation of a cloud of high-speed particles in the vicinity of the Moon for an extended period [5]. Importantly, these particles are not subject to slowing down, as the lack of atmosphere means there is no aerodynamic drag acting on them. This absence of drag significantly increases the duration of particle motion. In order to obtain a good understanding of the problematic, the following subsections will explain the characteristics of the lunar environment, lunar regolith, the interaction between the rocket plume and the lunar surface, and the review of previous studies that proposed and analyzed various designs of LLPs.

A. The Lunar Environment

The Moon's atmosphere bears a resemblance to the upper reaches of Earth's neutral atmosphere, known as the exosphere. Characterized by an extremely low gas density of approximately 100 molecules per cubic centimeter, the lunar surface has been scrutinized by detectors left behind by Apollo astronauts, revealing the presence of gases such as Argon-40, Helium-4, oxygen, Methane, Nitrogen, Carbon Monoxide, Carbon Dioxide, and, more recently, traces of water vapor from the Chandrayaan-1 mission [6]. At ground level, the atmospheric pressure on the Moon is notably present, measuring a pressure around 6.6×10^{-5} Pa and maintaining consistency across its entire surface.

Beyond the immediate lunar surface, the Moon's exosphere is a unique domain composed of particles that seldom collide. With an estimated total mass of approximately 25 metric tons, the

Moon's exosphere has not remained untouched by human activity. Each Apollo mission is believed to have introduced nearly 10 metric tons of gases into the lunar exosphere through engine exhaust alone [7]. This has raised concerns regarding the potential long-term impacts of vigorous lunar activities on the exosphere or the prospect of creating a sustained lunar atmosphere. The formation of such an atmosphere could disrupt scientific objectives and specific industrial processes that depend on the Moon's unique high-vacuum surface environment.

The Moon's poor atmosphere offers a minimal solar radiation protection. The lunar regolith experiences significant thermal fluctuations during a single cycle, with temperatures ranging from 100°C to -180°C [3]. In addition, studies on geopolymers created on Earth suggest that the mechanical characteristics of regolith remain relatively stable throughout these thermal cycles, making it an ideal material for various applications.

Furthermore, the atmosphere on the Moon influences the behavior of rocket plumes, differing from what has been studied on the surface of Mars, where the atmosphere will cause the rocket exhaust to collimate. This will likely cause much deeper craters to be produced than those during the Apollo missions on the Moon [8].

B. Lunar Regolith

The lunar regolith is the upper layer of lunar soil with a depth ranging from 4 to 15 meters and is roughly categorized, depending on the geological composition, into two types: highlands and mare [9]. Highlands consist of mountains made of lighter materials, such as anorthosite, formed by slowly cooled lava with a thickness of 10 to 15 meters. Conversely, mare refers to flat, dark areas formed by large lakes of ancient basaltic lava, typically with a thickness of 4 to 5 meters. This regolith results from constant meteoric bombardment and agglutination, potentially fragmenting rocks, basalts, glass, and other minerals.

In addition, the Moon's surface is enveloped in a cloud of dust that is generated by small particles from comets impacting the lunar surface. Lunar regolith, deeper than a few centimeters, is highly compacted and the absence of an atmosphere prevents abrupt pressure gradients, during the plume surface interaction, that could deform the soil [10].

Describing the layer of particles on the Moon's surface generated by meteoric impacts, lunar regolith is similar to volcanic ash. The finest component is what is known as dust ($<100\mu\text{m}$). The average regolith grain size is approximately $70\mu\text{m}$, with 10 to 20% by weight being $<20\mu\text{m}$ [11]. To put it on perspective, the average regolith grain size is too small to be observed with the human eye. These grains exhibit highly variable shapes, from spherical to extremely angular, although they are mostly elongated.

The lunar surface has experienced dust transport mechanisms closely correlated to the regolith composition. The lunar dust deposition rate induced by micro meteoroid impacts was estimated to be around $10\mu\text{g}/\text{cm}^2/\text{year}$ [12], while electrostatic transport could cause a much higher deposition rate of up to $3 \times 10^4\mu\text{g}/\text{cm}^2/\text{year}$ [13]. Recent research based on Apollo Dust Detector Experiments data suggests an upper limit of $100\mu\text{g}/\text{cm}^2/\text{year}$ for lunar dust deposition induced by both natural and anthropogenic causes.

During Lunar Roving Vehicle (LRV) excursions in the Apollo Moon landings, dust became a significant issue. It caused increased friction at mechanical surfaces, leading to wear and tear that significantly limited the lifetime of surface equipment. The abrasive nature of lunar dust is quite challenging as well as its ease to stick to the surfaces. A good example of this problem can be described during the Apollo missions, where the astronauts covered their suits with dust during the Moon walks, therefore introducing the adhered dust into the spacecraft. The astronauts reported the presence of dust and respiratory problems after the exposure. Specifically, the astronaut Gene Cernan reported these problems [14].

Moreover, the performance of electronic and thermal control devices were degraded, as detected during Lunar Roving Vehicle (LRV) excursions [11]. It is possible to summarize the dust-related hazards falling into nine categories: vision obscuring, false instrument readings, dust coating and contamination, loss of traction, clogging of mechanisms, abrasion, thermal control problems, seal failures, and inhalation and irritation.

Notably, spacecraft launches emerge as a significant factor in local lunar dust transport from the source. Despite the challenges posed by lunar dust, the actual deposition rates observed during Apollo's explorations in the northern Mare Imbrium were within a manageable range,

highlighting the importance of ongoing research and understanding these lunar surface dynamics.

C. The Plume-Surface Interaction

As it is stated before, the interaction between the plume of a rocket and the Moon's surface is crucial in the development of a mission. Therefore, it is important to understand the way this interaction occurs and what are the mechanisms of erosion of the Moon's surface, in order to develop an efficient mitigation system that prevents the formation of dust clouds and sandblasting.

The effects of the Plume-Surface Interactions (PSI) are highly dependant on the gravity, atmospheric density, mass and thrust of the rocket, properties of the surface, engine configuration, etc. In the case of the Moon, there is an almost non-existing atmosphere and its gravity is relatively low in comparison to the one on Earth. Furthermore, the Moon surface is covered by dust and rocks, that are blown away at high velocities.

During lunar landings, the PSI generally starts when the spacecraft is around 30 or 40 meters above the lunar surface [15]. However, the plume effects go beyond what is visually observed. The interaction between the plume and the Moon starts even before the spacecraft reaches the previously mentioned altitudes, where the exhaust gases are being released into the lunar exosphere. In addition, as the rocket exhaust species usually contain H₂O and NH₃ (species that also exist at the lunar poles [16]), the molecular concentration in the exosphere and the regolith increases after each engine use.

Once the rocket is approaching the surface, a stagnation region is formed directly under the nozzle and the favorable pressure gradient turns the flow outwards, hence creating a high velocity jet tangential to the surface [11]. Thus, the particles near the center are printed with a higher vertical speed and, therefore, reaching higher altitudes, whereas the particles at a distance of 2 or 3 meters from the center have a high horizontal speed [9]. This zone is the most critical, where high amounts of dust are accelerated at velocities reaching up to 3km/s [15], in many cases exceeding the escape velocity of the Moon (2.4km/s) and even inserting the dust into lunar orbit and impacting orbital hardware.

The hazards that represent the lunar dust ejecta require changes in instrument mounting locations, installation of shields and hardened covers or

deployment mechanisms. In the case of multi-engine landers, the risk increases because the plume can redirect hot gas and ejecta blast back up towards the vehicle [15]. Generally, the horizontally ejected particles travel close to the surface in angles of 1 to 3 degree with respect to the surface, additionally, the local craters eject streams of dust into higher angles and, during the final moments of landing, the plume ejects particles near the center into higher angles (around 15 degrees) [10]. In conclusion, there is evidence that demonstrates that larger particles, traveling at high speeds, eject at a lower angle of 3 degrees, whereas slower and smaller particles eject at a higher angle closer to 17 degrees [17].

Regarding the lunar surface, it is important to highlight the three identified erosion mechanisms that arise from the interaction with the rocket plume:

- Viscous Erosion: It is the most dominant mechanism and occurs when the shear stress from the rocket plume exceeds the shear strength of the soil. The shear strength is determined by the Mohr-Coulomb Equation [18].

$$\tau = c + \sigma \tan \varphi \quad (1)$$

Where τ is the shear stress, c is the soil cohesion, σ is the pressure normal to the surface and φ is the angle of internal friction.

- Bearing Capacity Failure: Occurs when the pressure over the soil exceeds the bearing capacity of the soil and a narrow cup is formed. The walls of the narrow cup are unstable and collapse, releasing many dust grains into the flow field while forming a crater.
- Diffused Gas Eruption: It is carried out when the gases coming from the rocket get through the pore spaces of the soil and, eventually, the gases will come out, bringing particles along.

On the Moon's surface, the parameter σ is expected to be relatively low, indicating that the soil cohesion plays a significant role in the viscous erosion of the regolith. The specific value of σ can vary based on the in-situ properties of the regolith at a given lunar location, but it is generally estimated to fall within the range of 100 to 1,000 Pa [19]. Future Artemis class landers are anticipated to incorporate even more powerful engines, amplifying the shear force exerted during upcoming missions. In addition to the shear force, the temperatures in the region affected by the

rocket plume are projected to be in the range of 3000-4000 °C at the point of impact.

According to these mechanisms, soil erosion has been observed during many lunar missions. One of the best examples is the Apollo LM descent (Figure 1) [20] video footage, where regolith erosion could be observed. Further analysis of the images led to the estimation that, apporximately, 2.5 tons of material were ejected by the 5-ton LM during landing.



Figure 1: Apollo 15 Landing Footage [20].

Similar analysis has been performed on footage from China's 0.8-ton Chang'e 4 lander, providing an estimation of 19 kg of ejected material. These data led to the rough estimation that the amount of ejected material scales as the vehicle mass to the 2.5 power [10] in lunar environment, whereas at 1 atm it seems to scale linearly. This relationship is shown in Equation 2[18], where dm/dt is the rate that soil mass enters the gas flow per unit area and σ is the shear stress on the soil at every location $\vec{R}$ and time t .

$$\frac{dm}{dt} = 0.222\sigma^{2.52}(\vec{R}, t) \frac{kg/s}{m^2} \quad (2)$$

According to Burns et al. [18], this equation gives a good prediction of regolith ejecta for Artemis Moon Landing, the next mission that will send humans to the Moon. Although the exact mass of the Human Landindg System (HLS) that will be deployed is unknown, NASA indicated that it will be around 40 tons, giving us an approximation of 470 tons of ejected material.

As it is stated before, this amount of ejected material is very dangerous for the structures in the surroundings and even the lander itself. There are two good examples that were analyzed in order to determine the effects of sandblasting, Surveyor III in the Moon and Curiosity in Mars.

Surveyor III: The pieces of Surveyor III provide the best evidence of rocket exhaust plume damage that can occur during landing. Apollo 12 landed approximately 155 meters away of Surveyor III and the astronauts gathered parts for its examination on Earth. It was observed that the surfaces experienced pitting and cracking [15] as a result of exposure to

blowing dust. The Surveyor III was located in a crater slightly below the main spray of dust from the Apollo 12 exhaust plume. If the Surveyor III was exposed to the direct stream, the damage would have been orders of magnitude greater [21].

In more than 36 studies of the Surveyor III parts conducted by more than 80 investigators, almost all of the exposed surfaces were at least partially covered with a layer of lunar dust. The dust distribution along the parts indicated that the lunar dust was disturbed and implanted both upon the initial landing of Surveyor III and the landing of the Apollo 12 Lunar Module [7].

Curiosity: During the landing of MSL in 2012, pebbles were blown during the rocket-powered landing, striking and damaging one of the Curiosity rover's two wind sensors [22]. Plume-Surface Interactions were present since 63 meters above the ground, and erosion continuously increased as the vehicle descended. At around 6 meters above the surface, large particles of soil were blown up, impacting the bottom of the rover chassis. This produced permanent damage on the wires of the wind sensor [15].

Notably, it is important to highlight that the lunar lander designs under development include the capability of decreasing engine thrust or even shut them down at a certain altitude to reduce the PSI. Therefore, it is assumed that the worst case scenario is the take off of the vehicle from the lunar surface [15], where the engines are at its closest point at a relatively high thrust.

In conclusion, there is a need of mitigating the PSI as much as possible to prevent damage and failure of hardware or any other structure operating in the Moon. One of the most studied solutions for mitigating PSI is the creation of Lunar Landing Pads (LLP), where the main challenge is prevent the plume gas from flowing through the cracks between the pavers, that could form a large area of high-pressure gas under the pad [5], resulting in a catastrophic failure. The upcoming section will go through some of the previously proposed mitigation strategies and various LLP designs.

D. Mitigation Ideas

Many studies have been performed in order to address possible solutions to the lunar dust problematic. Different approaches have been considered in order to reach to an efficient solution or set of solution candidates to be

implemented in the future Moon exploration missions.

The most suitable and most commonly studied approach is building a Lunar Landing Pad (LLP) with in situ resources. However, there exist also other smart approaches that have been considered and widely studied, although, some of these are not able to solve the problematic on its own. These approaches are:

- **Propellant Optimization:** According to some studies, there is evidence that some propellants may be more suitable than others where contamination is a concern. Additionally, some propellants could slightly reduce the Plume-Surface Interaction.
- **Sprayer Landing Pad:** This approach intends to build the LLP during the hovering phase using spray technique. This method uses engineered, generally Alumina-based, particles injected into the rocket plume to build up a coating over the regolith at the landing location. The hardened regolith would have greater thermal resistance and ablation resistance to reduce regolith erosion rates and deep cratering [23]. The process is represented in Figure 2. This technique could be applied to any lander size and avoids transporting heavy loads to build the landing pad on the surface. However, several factors should be taken into account, such as melting point of the material, the risk of clogging the engine, etc.

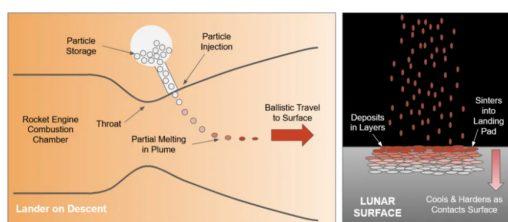


Figure 2: Particle injection in the engine (left) and deposition on the surface (right) [24].

- **Deployable Landing Pad:** This idea consists in dropping the pad from the module at a certain hovering altitude or in the previous cycle. Once it reaches the surface, the platform is deployed. Its main purpose is to absorb the kinetic energy from the gas ejected by the nozzle, avoiding cratering and reducing PSI. It is estimated that the pad should have around 8 meters in diameter at least [9]. The main constraint of this approach is the need of light materials that

could sustain the thermal shock from the exhaust gases. Furthermore, the precision on the drop is very crucial and limits its effectiveness.

Regarding the Lunar Landing Pad construction using in-situ resources, many authors have identified and considered a two-zone strategy for LLP, since the requirements for the pad material are different in each zone.

The first zone is in charge of withstand the high temperature and pressure of the shock event at the engine ignition for launch and during landing. It also deflects the plume coming from the nozzle, transforming it into horizontal flow that comes into the second zone. The region of this inner zone is limited to a few meters around the engine nozzle, due to the radial expansion of the gases from the center point, dropping in temperature and pressure.

Despite the inner zone is thought to be prepared to withstand the pressure and temperature, the surrounding areas can be eroded by the radially expanding gases and may eject soil. Therefore, the second (outer) zone must be designed to stop erosion along a larger radius.

Previous work from van Susante and Metzger [25] have estimated a first zone radius of 5 meters and a second zone radius of 20 meters in order to target the plume of a 40 tons lander. Notably, this numbers are estimations and not necessarily close to reality. Many authors have considered this as a starting point for the size estimation of their designs.

Autry and Metzger [5] added 7 meters to each radius in order to account for a multi engine lander that has engines 2 meters diagonally off center line and a traditional 5 meter margin in the landing accuracy. The design of this LLP is represented in Figure 3, where they even proposed materials and construction methods.

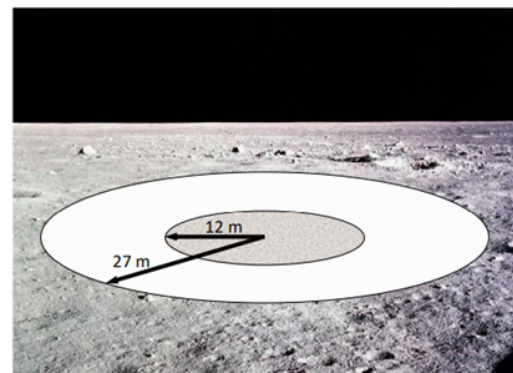


Figure 3: Autry and Metzger LLP design [5].

The inner zone could be made of polymer infused regolith but it would require maintenance after each landing to patch the ablated material. The outer zone could also be built with polymer infused regolith, but without the cost of requiring constant maintenance after use. The temperatures will be lower and the soil only needs to resist the shear stress of the gas, but it introduces the challenge of bringing tons of polymer additive from Earth.

Utilization of in-situ materials by microwave sintering seems to be especially efficient since the lunar dust contains nanophase iron (np-FeO) particles coating the grains [5]. Microwave sintering is attractive because it is a simple method and do not require complicated robotics to be carried out, moreover, the long wavelength of the microwaves produces good penetration in lunar soil, hence, producing an enough competent sintered material after a single pass.

A different LLP design approach was proposed by a research group at the Colorado School of Mines [18]. This design, called LILL-E Pad, uses a different approach for the inner region of the landing pad that is the utilization of a 25 meter radius carbon fiber blanket that will be anchored to the surface of the Moon, whereas the outer region is a 100 meter radius layer of regolith stabilized using ultraviolet (UV) cured resin, as seen in Figure 4.

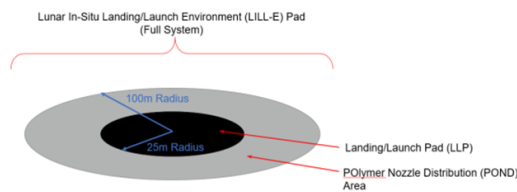


Figure 4: LILL-E Pad System Schematic [18].

The first element to be deployed is the resin to build the outer region. The carbon fiber blanket is deployed on top of the stabilized regolith and it is intended to resist the most extreme area of the rocket plume at the same time as it inhibits gas flow to the lunar surface. However, this has to be brought from Earth and is also subject to deployment accuracy constraints.

Another interesting LLP design was proposed by El Kraye, Drago et al. [9]. In this approach, the LLP takes into account three different regions, each one for one specific issue, as it is explained below and observed in Figure 5.

The first and central region consists in a small compact hill, made of melted regolith, the zone where regolith usually spreads with higher vertical velocity, as seen in Subsection II-C. This hill also redirects the outgoing gases from the nozzle, smoothing the impact of the plume on the surface with its geometry. This region has a radius of 4.5 meters and a height of 0.5 meters.

The second region extends on the surroundings of the hill. It is a flat area composed by three layers of stones made of melted regolith and natural rocks found in the area. The purpose of this region is to reduce the spreading of the dust and fix the ground below. This region has a radius of 10 meters.

Finally, the third region intends to isolate the external area by building a semi-circular bank that completely surrounds the other two regions. It has a height of 1.5 meters and is useful to capture the spread regolith from the inner regions and, in addition, to rise the angle of the outgoing flow.

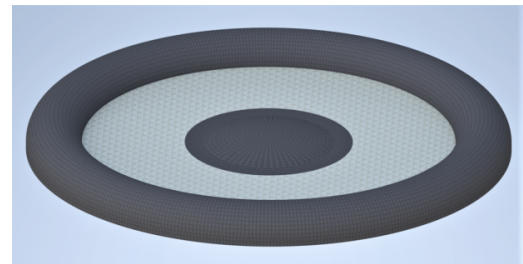


Figure 5: LLP design for PSI mitigation [9].

This landing pad keeps the PSI phenomena restricted to a region of 13 meters radius, leaving the dust around almost unaltered and keeping the lander safe due to the central hill of compact soil. The construction method considered for this design is the microwave sintering, a very suitable technology that has already been developed and tested.

The last two studied designs that will be reviewed in this document are proposed by Campbell et al. [26].

The first design is quite complex, having many parts, as it can be observed in Figure 6. The surface of the external part has grates to capture the exhaust gas in the vertical direction, however, it does not completely prevent the backflow of the plume. The grates allow the exhaust to enter inside of the pad, where the flow will be redirected by the rest of the parts.

The central cone redirects the exhaust flow from the center of the pad in order to expand it radially,

additionally, it also increases the structural integrity of the pad.

Surrounding the central cone, there are vent dividers, that assist in directing the exhaust gas from the center of the pad to what is called the KEDs. Depending on the shape, there are two different types of vent dividers: the teardrop and the leaflet. The rounded side of the outer teardrop vent divider allows more negative space for exhaust to flow into the high-pressure converging point before flowing out of the KEDs. Each of the inner leaflet and outer teardrop shaped vent dividers prioritizes directing flow to the KEDs and avoiding high-pressure regions [26].

Finally, in the external region of the LLP, there are the previously mentioned KEDs (Kinetic Energy Diffusers). These KEDs are openings to let the gases out in a relatively controlled way. These openings are observed in the Figure below.

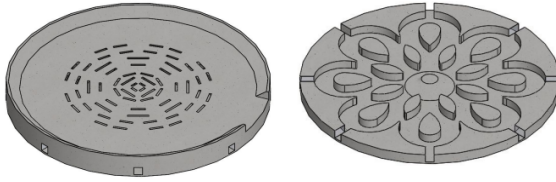


Figure 6: LLP design full view (left) and internal view (right) [26].

As an improvement to this design, the authors have hypothesized a full-scale version, that is more complex and complete than the one showed in Figure 6. This full-scale concept include deflectors at the exit of KEDs, ramp, ramp deflector, etc. As observed in Figure 7 below.

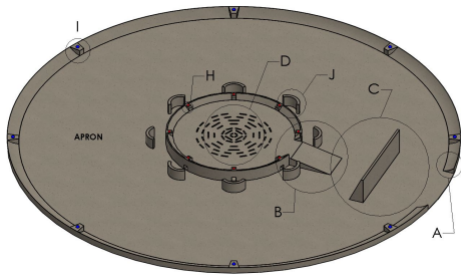


Figure 7: LLP design full-scale concept [26].

In addition to a larger pad diameter, a dust-protective apron extending to approximately twice the pad diameter is recommended, as well as an additional deflector to contain the plume at the perimeter [26].

Finally, the last design proposed by Campbell et al. [26] consists in a blast shield at the perimeter of a flat surface with the aim of mitigating the high speed particles, that move at shallow angles (between 3 and 17 degrees as seen in Subsection II-C), by deflecting them to a concave zone, as shown in Figure 8. This blast shield would have a height of approximately the tangent of three degrees (supersonic ejecta) to seventeen degrees (subsonic ejecta) times the pad diameter to mitigate the effects of the high-speed plume and particles [26]. A smaller diameter pad could reduce the required height for this blast wall.

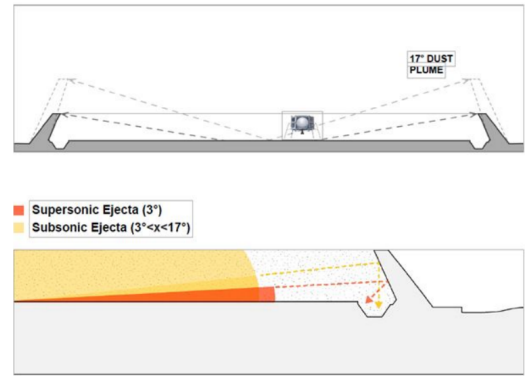


Figure 8: LLP blast shield consideration to mitigate sandblasting [26].

To conclude, the designs that have been reviewed here, are different approaches to address the same problematic. All of them try to prevent erosion, by using different materials, with the aim of avoid new material being accelerated by the plume, joining the flow and increasing the sandblasting. But, the recent designs, have gone one step further and have considered the exhaust gases flow control as a promising option either by increasing the angle of ejecta or by attempting to confine the gases and ejected material, in addition to the erosion prevention ideas. It has also been concluded that the most suitable method of construction is the microwave sintering of lunar regolith.

III. METHODOLOGY

The methodology of this project starts with an exploration of the software employed in both design and analysis, subsequently, the procedural workflow will be explained, going through the design and through the various steps of analysis, justifying boundary conditions and model data.

A. Software Tools

The two software tools used in this project are Fusion 360 for the 3D design part and Cradle CFD for the fluid dynamics analysis part. A quick, further explanation of these two tools is exposed below.

1) Fusion 360 (Design): Fusion 360 is a software developed by Autodesk and it is a cloud-based CAD/CAM/CAE tool for collaborative product development. It combines quick and easy organic modeling with precise solid modeling to create designs that are easy to manufacture [27]. It is also a very intuitive and versatile design software, easy to learn and good for iteration procedures.

2) Cradle CFD (Analysis): Cradle CFD is a multiphysics-focused computational fluid dynamics software owned by Hexagon, that can be obtained in the same package as other softwares like MSC Patran or MSC Apex. It consists in many modules but, in particular, in this project only the module scFLOW will be used, due to its convenience for CFD analysis.

ScFLOW is a next generation CFD tool that uses unstructured mesh to accurately represent complicated geometry. With Preprocessor that helps any level users generate high quality polyhedral mesh elements and construct complicated models, and Solver that ensures more stability and speed, scFLOW is capable of solving aerospace and automotive aerodynamics, performance of fans, pumps and other rotating equipment, design problems of electronic devices, multiphase phenomena, marine propeller cavitation, and varied problems [28].

B. 3D Design: Scope And Purpose

The first step to create a 3D Model of a Lunar Landing Pad is to understand the problem to be addressed by the design and to know what is the size of the rockets that will land or take off. As it is showed in Section II, the possible approaches of the designs are, on one hand, focusing on preventing the erosion of the lunar soil, aiming for a simple design, or, on the other hand, focusing on confining and deflecting the flow coming from the plume, also implementing the erosion prevention, leading to more complex designs.

The name chosen for the LLP design is COOL-PAD (Confiner On-site Operational Landing PAD) and the purpose of the design developed in this project is to confine and increase the angle of ejecta from the flow by redistributing

it along the landing pad. This way, the particle acceleration above lunar escape velocity and the sandblasting phenomena are mitigated. This design is also aiming to withstand the Starship HLS (Human Landing System), that is the spacecraft selected by NASA to land on the Moon in the next Artemis manned mission. Therefore, the sizing of the COOL-PAD has to be enough to target the plume of various Raptor rocket engines, that will compose the Starship HLS.

Taking inspiration from the designs showed in Figures 5 & 6, the design concept is composed of various parts. Most of them are common for all the different versions of the COOL-PAD, but, it is possible to classify into three categories:

- **Center Hill:** The hill in the center of the platform is an idea inspired by El Kraye, Drago et al. [9]. Its purpose is to smooth the first interaction between the plume and the surface by redirecting the flow to the horizontal region with a better, less abrupt transition. The size of the hill is determined by the diameter of the Starship HLS, which is of 9 meters. Therefore, the diameter of the hill will be of 9 meters at least, with a height of 0.5 meters, so it covers the plume or, at least, most of it, while it allows the spacecraft to land on a horizontal surface.
- **No-man's Land:** This part is a flat region between the center hill and the confiner walls. The size of the region depends on the version of the COOL-PAD, because the confiner walls differ in shape and/or location.
- **Confiner Walls:** This is the most differing part between the versions of the design obtained along the iterations. It goes from V shape in the COOL-PAD v1, as observed in Figure 9, to C shape in combination to O shape in the latest versions, similar to what is observed in Figure 6, in fact, it served as inspiration for the latest iterations.

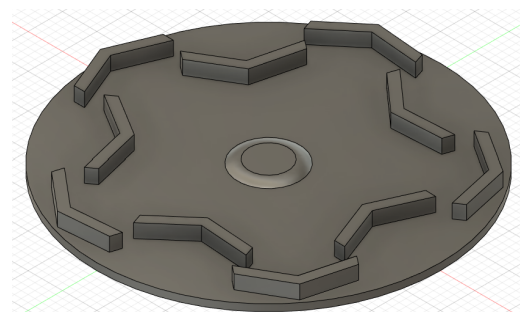


Figure 9: COOL-PAD version 1 3D Model in Fusion 360.

It is also considered for the designs to include free space between the confining walls and never close completely the platform. Therefore, during a hypothetical manned mission, the astronauts are able to exit the platform walking once they land on it.

In order to design the 3D Model with Fusion 360, the base platform of the landing pad is created with a cylinder with a radius of 25 meters, one of the base geometries of the software. The hill is created with another cylinder, in contact with the previous one, and its curvature is performed by using the fillet tool. For example, in the COOL-PAD v1 (Figure 9), the fillet has a radius of 3 meters, whereas in the next versions, the fillet has been adapted to cover the whole hill, therefore fully generating curvature on its surface.

The Confiner Walls have been created using the Sketch Tool within Fusion 360. First of all, an original wall patron is created, as observed in Figure 10 below, that will be copied and pasted using the rotation option of the Sketch Tool. The amount of copies created by rotation varies on the version, for instance, in this COOL-PAD v1, there are four rotations by angles of 72 degrees, in order to create five copies of the same walls.

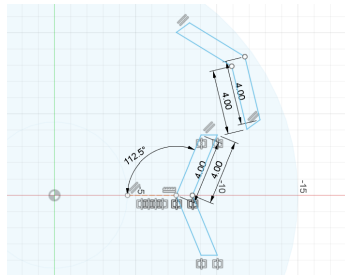


Figure 10: COOL-PAD v1 Confiner Walls original drawing in Fusion 360 Sketch Tool.

Some of the next COOL-PAD versions have up to six and eight rotations of 60 degrees and 45 degrees, respectively.

C. Analysis: Procedure, Boundary Conditions and Load Cases

As it is stated before, the purpose of the analysis in Cradle CFD is to create a Finite Element Model (FE-Model) of the 3D Model with the aim of performing a fluid dynamics analysis in order to simulate the flow of exhaust gases during landing.

The FE-Model is a discretization of a continuous domain that is subdivided into reference elements

and nodes, known as approximation with finite elements. These reference elements are transformed into continuous elements by constructing interpolation functions, characterized by the Jacobian matrix [29]. Now, it is important to know the procedure for creating the FE-Model for its further analysis.

Once the 3D Model is exported from Fusion 360 in STEP format (.stp), subsequently, it is imported in Cradle CFD, which creates a product tree that gathers all the different bodies from the model, it means, if the 3D Model is composed in just a single body, there will be only one part in the product tree.

When the model is imported in Cradle CFD, the first step before start working with it, is to create the Computational Domain. The Computational Domain is necessary in order to restrict the calculation to a finite region. The typical measurements in Cradle CFD when performing an aerodynamic analysis are 2L for upstream direction, 10L for downstream direction and 5L for vertical and horizontal directions [30]. Due to computational limited resources, for this analysis, the Computational Domain will be a cylinder with a diameter of 150 meters, which is 3 times the COOL-PAD diameter, instead of 5 times, as it is stated before, and a height of 15 meters, which will be the height of the plume incising onto the platform. The Nozzle is another cylinder on top of the Computational Domain, with a diameter of 9 meters (diameter of the HLS propulsion system) and a height of 1 meter, as it is observed in the Figure 11 below. However, the height of the Nozzle cylinder is not relevant.

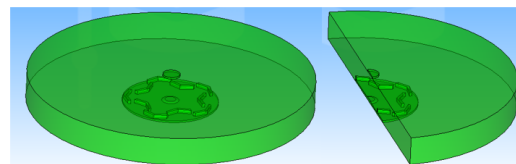


Figure 11: COOL-PAD version 1 Cradle Model with Computational Domain and Nozzle. Full Model (left) and Half Model (right).

After all necessary parts have been created, in order to reduce the computation cost in time and resources, the model is cut in half, as observed in the previous Figure, with the purpose to impose a symmetry condition in the resultant face of the cut. This method will reduce the computational cost without threatening the reliability of the simulation.

Once the geometry processing has been done, the model is prepared to start the simulation process. First of all, it is necessary to create the Octree. The Octree is a method of dividing three-dimensional space repeatedly into eight division, where each divided space is called an octant [31]. This process is necessary for the generation of unstructured mesh by creating groups of cuboids with Octree and adjusting mesh scale and coarseness.

When the Octree is generated, it is time for meshing the model. The meshing option allows to generate automatic unstructured mesh. Unstructured mesh is a mesh where triangle or rectangle (for two dimensional use), or polyhedral (for three dimensional use) mesh elements are irregularly aligned. Unstructured mesh can be allocated flexibly, which makes it possible to represent geometries accurately [32]. Since this is a three dimensional model, the mesh is polyhedral and, despite its generation is automatic, it can be customize by targeting the faces of the model and specifying a certain thickness coefficient and the number of layers over the selected surface. The Octree and mesh of the COOL-PAD v1 can be observed in Figure 12 below.

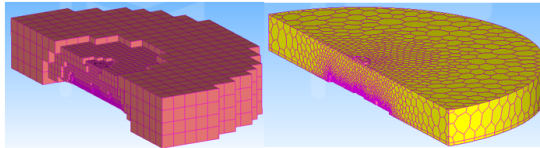


Figure 12: COOL-PAD version 1 Cradle half model. Octree (left) and Mesh (right).

Notably, the faces that must be specifically addressed by the mesh are the Central Hill and the Confiner Walls, where its shape has to be as similar as possible to the 3D Model in order to obtain reliable results. In the previous Figure it can be clearly seen that the closer it gets to the central part of the platform, the finer is the meshing.

Following, once the model is geometrically prepared for the analysis, it is necessary to apply the materials, the loads and impose the boundary conditions (BCs) in order to perform the flow analysis.

1) Materials: In order to determine the gases that compose the plume of the SpaceX HLS, a simulation with NASA's tool CAE (Chemical Equilibrium with Applications) [33] has been performed. Here, the rocket code type is selected

and the initial data is introduced. The data refers to SpaceX Raptor Engine, that will compose the propulsion system of the HLS. The properties of the Raptor Engine to be introduced in CAE tool are the following [34]:

- Chamber Pressure: 350 bar.
- Fuel: Liquid Methane (LCH4).
- Oxidizer: Liquid Oxygen (LO2).
- Oxid/Fuel ratio: 3.6.

Where it is obtained that the main exhaust species are CO₂ and H₂O. Additionally, the result of this simulation has been correlated with an analysis report of the Raptor Engine performed by the FAA (Federal Aviation Administration) of the United States of America. This report [35] concluded that the most dominant species among the exhaust gases are CO₂ and H₂O.

Then, the selected gas for the simulation is CO₂, because considering a mixture of gases in Cradle CFD would cause problems in the simulation due to the constraint in computational resources. In order to obtain the properties of the CO₂ at the exhaust temperature (2500 K [34]), and, interpolating data, an external chemical calculator [36] has been used. This data is presented in the following table:

Property	Value	Unit
C_p	1397.5 6	J/(kg.K)
C_v	1208.5	J/(kg.K)
R	189	J/(kg.K)
Density*	0.13	kg/m ³

Table I: CO₂ Properties at 2500 K.

Since Cradle does not allow to define inflow (m/s) and mass flow (kg/s) at the same time, the density in Table I does not correspond to the real value. This value has been adjusted to fit the mass flow of the Raptor Engine (see Equation 3) while maintaining the velocity of the exhaust gases (see Section III-C2). 650 kg/s corresponds to the mass flow of the Raptor Engine, 3700 m/s to the exhaust velocity of the gases, and 0.65 m to the Raptor Nozzle radius.

Additionally, the value of the Gas Constant R has been calculated from the constant pressure heat capacity C_p and the constant volume heat capacity C_v values, using the relation from Equation 4 [37].

$$\rho = \frac{650[kg/s]}{3700[m/s]\pi 0.65^2[m^2]} \quad (3)$$

$$Cp = Cv + R \quad (4)$$

Then, the material is applied to the Nozzle region and the Computational Domain region as an incompressible fluid, because gas mixture is deactivated and compressible analysis is very demanding in terms of computation power. In the following section, the conditions applied to emulate the vacuum environment will be exposed.

To conclude, the solid regions will remain in the model as obstacles with no specified material.

2) Boundary Conditions: The Boundary Conditions are responsible for the model characterization. The BCs imposed in this analysis are listed below:

- **Analysis Type:** It is a pure flow analysis, not taking into account heat, mixed gas or other type.
- **Initial Condition:** Here, the vacuum conditions will be approximated by applying an initial value for Pressure of 0.1 Pa. This is a very low value but it can not be introduced as 0 Pa because this will generate divergence in the simulation.
- **Flow Boundary:** The Inflow Condition is applied to the upper, circular face of the Nozzle region and it imposes an incoming flow of 3700 m/s, that corresponds to the exhaust velocity of the gases coming from the Raptor Engine. Furthermore, an Outflow Condition is applied to the wall of the Computational Domain region, imposing an Outflow Static Pressure of 0 Pa.
- **Wall Boundary:** There are two different conditions here. The roof of the Domain Region will have a Free-Slip Wall Stress condition, meaning that the surface is not confining the gases. The other condition is applied to the Central Hill, the Platform and the Confiner Walls, introducing a No-Slip Wall Stress condition with an equivalent roughness of 0.005 meters [38]. The equivalent roughness of the lunar regolith varies from the scale of millimeters to centimeters.
- **Symmetrical Boundary:** This symmetric condition is applied to the cross section surface resulting from the cut performed during the preprocessing.

3) Load Cases: The Load Cases selected for this type of analysis are the steady-state and transient analysis. The steady-state approximates time to infinite and tries to find the equilibrium and convergence after a certain number of iterations. The transient analysis is characterized for being finite in time and having time steps (cycles), this way, it is possible to define a time step and a number of steps to carry out a simulation during a finite time. The drawback of this analysis compared to the steady-state is that it is way more time-consuming, but it is more close to reality.

Regarding the transient analysis, it is a matter of interest to simulate the plume flow impacting on the COOL-PAD for a duration of 15 seconds, similar to [14]. The time step selected is 0.075 seconds, thus, completing the analysis in 200 cycles. This time step has been selected because it provides a reasonable computation time without risking the accuracy of the model.

To summarize, the steady-state and the transient analysis will consist of an incoming flow of 3700 m/s with a mass flow of 650 kg/s (tuned in the CO₂ density), from an altitude of 15 meters during 15 seconds in 200 steps (0.075 seconds each).

4) Solver: The solver uses different sets of equations depending on the type of analysis to be computed in the model:

Flow	U, V, W : momentum equations P : mass conservation (pressure correction)
Temperature	T : energy conservation
Turbulence	K : k equation E : ϵ equation

Figure 13: Cradle CFD scFLOW solver sets of equations [39].

It is possible to obtain velocity field from momentum conservation equations, pressure field from mass conservation (coupled with the momentum equations), and temperature from energy conservation. It is also possible to use the k - ϵ equations for turbulence energy k and dissipation rate ϵ that represent the intensity of turbulence and the rate of energy dissipation, respectively [39]. Additionally, the continuity equation, that specifies the mass balance of flow, is included to fulfill the governing equations in thermofluid analyses.

The finite volume method is used in the solver to compute the mentioned equations. A finite volume method solves the governing equations by converting them into an integral conservation form that is expressed on each fractional unit of decomposed elements, or control volume [39]. Both flow velocities and scalar quantities are defined on the nodes.

As shown in Figure 14, two control volume formulations are widely used with the finite volume method [39]. The cell-centered scheme utilizes a decomposed mesh element itself as a control volume. On the other hand, the vertex-based scheme encloses a node point in a control volume by connecting the centroid of each element and the medians of edges. ScFLOW uses a vertex-based scheme, since the discrete equations are written in a relatively simple form, and also the number of control volumes reduces compared to cell-centered formulation in a tetrahedral mesh.

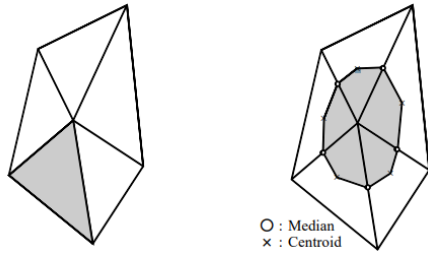


Figure 14: Control volume formulation. Cell-centered scheme (left). Vertex-based scheme (right). [39].

In the numerical analysis, the Computational Domain is decomposed with many elements. Using a finite element method, the conservation law is solved on each control volume [39]. By applying it to the adjoining elements, a chain of inflow/outflow relations can be formed (outgoing budget of element A is the incoming for B and so on). Mathematically, this is equivalent to solving the system of equations, or seeking an inverse of the matrix that expresses the system. Therefore, by numerically solving the matrix equation on a computer, we can reach the answer to satisfy the balance of all the budgets for each control volume [39].

The equations used in scFLOW include mass conservation, momentum conservation, energy conservation, turbulent equations, and conservation of diffusive species for both compressive and incompressible fluids. The

equations of interest are the following [39], only for incompressible fluids:

- Mass Conservation Equation:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (5)$$

- Momentum Conservation Equations (i=1-3):

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial u_j \rho u_i}{\partial x_j} = \frac{\partial p}{\partial x_i} \quad (6)$$

$$\frac{\partial}{\partial x_j} \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) = \rho g_i \beta (T - T_0) \quad (7)$$

- Energy Conservation Equation:

$$\frac{\partial \rho C_p T}{\partial t} + \frac{\partial u_j \rho C_p T}{\partial x_j} = \frac{\partial}{\partial x_j} k \frac{\partial T}{\partial x_j} + \dot{q} \quad (8)$$

- Equations of Turbulent Energy and Turbulent Dissipation Rate (k - ϵ equations):

$$\frac{\partial \rho k}{\partial t} + \frac{\partial u_i \rho k}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{\mu_t}{\sigma_k} \frac{\partial k}{\partial x_i} \right) + G_s + G_T - \rho \epsilon \quad (9)$$

$$\frac{\partial \rho \epsilon}{\partial t} + \frac{\partial u_i \rho \epsilon}{\partial x_i} - \frac{\partial}{\partial x_i} \left(\frac{\mu_t}{\sigma_\epsilon} \frac{\partial \epsilon}{\partial x_i} \right) = 0 \quad (10)$$

$$C_1 \frac{\epsilon}{k} (G_s + G_T) (1 + C_3 R_f) - C_2 \frac{\rho \epsilon^2}{k} = 0 \quad (11)$$

Where the meaning of the variables are:

x_i = Coordinates [m].

u_i = Velocity of flow in x_i -direction [m/s].

t = time [s].

ρ = Density [kg/m³].

p = Pressure [Pa].

μ = Viscosity [Pa.s].

σ_{ij} = Stress tensor

H = Specific enthalpy [J/kg].

g_i = Gravity [m/s²].

β = Coefficient of volume expansion [1/K].

T = Temperature [K].

T_0 = Reference temperature [K].

C_p = Specific heat at constant pressure [J/(kg.K)].

K = Thermal conductivity [W/(m.K)].

$\dot{q}$ = Heat source [W/m³].

k = Turbulent energy [m^2/s^2].

ε = Turbulent dissipation rate [m^2/s^3].

When the model is prepared, it is time to launch the solver. The Cradle CFD solver use two types of files for computation. One is the GPH file (.gph) that contains the mesh data of the model and, the other is the SPH file (.sph) that contains the conditions of the analysis.

When the solver of scFLOW (scFLOWsolver) begins an execution, it first reads the SPH file containing the names of input and output files and solver conditions. An SPH file can be created or edited using the preprocessor of scFLOW (scFLOWpre) and consists of headers, file specification data, initial settings and commands [40]. Since it is a text file, users can also directly edit it with a text editor. In fact, for this project, and since this is a Cradle CFD Student Edition with limited computational resources, it was necessary to include some BUND command lines in the SPH text file to enlarge the element stabilization criteria, preventing the analysis from diverging. This command lines can be seen in Appendix A.

While the solver is running, it shows in real time the selected values to track (pressure, temperature, kinetic energy ...) as well as the convergence of all these variables during every computed cycle, as observed in Figure 15.

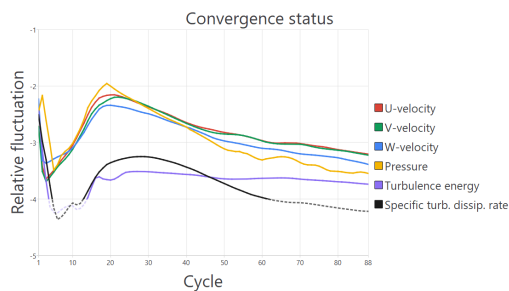


Figure 15: Cradle CFD scFlow Solver convergence window.

IV. RESULTS AND DISCUSSION

The objectives of this section are to explain the different configurations of the versions of the COOL-PAD (versions 1 to 5) and determine its performance by studying the results of the flow analyses, scrutinizing the aerodynamic characteristics of these designs, aiming to find the best configuration, among the ones proposed, to withstand and confine the exhaust gases released during descent phase.

In addition to this, finding the design parameters that contribute most to the efficiency of the lunar landing pad is also a matter of interest. The subsequent discussion will examine these findings, drawing connections to existing literature, identifying innovative design considerations, and contemplating the broader implications for future lunar exploration missions.

The results are obtained using the postprocessor of scFLOW (scFLOWpost) already incorporated in Cradle CFD software and the results are split into steady-state and transient and, in addition to this division, structured according to each version of the COOL-PAD, with a brief explanation of the design and changes with respect to the previous version, with the exception of versions 4 and 5, which are more sophisticated designs and do not follow the evolution line of the versions 1, 2 and 3. After all the results are exposed, the discussion section will gather them for further interpretation.

A. Results: Steady-State Analysis

As it is specified before, the steady-state results are going to be presented first, for each version of the COOL-PAD, and they will include the pressure plot, the wall shear stress plot and the flow velocity plot.

1) COOL-PAD version 1: This first version of the landing pad corresponds to the 3D Model shown in Figure 9 in Section III-B. This design has a Central Hill with a diameter of 9 meters, with a flat circle in the middle with a diameter of 5.6 meters. Furthermore, the platform has a diameter of 50 meters in total and has 10 V-shape walls, 5 looking inwards and 5 looking outwards, each one spread by an angle of 72 degrees from the consequent of the same type. The height of the V-shape walls is 2 meters and its thickness is 1.4 meters.

The pressure results are represented in the Figure 16 above and it provides a maximum pressure of around 1 MPa, distributed along the inner part of the Central Hill. It is also observed that the pressure is high, at the edge of the hill, in comparison to the platform, and it is also high at the bottom edge of the V-shape walls.

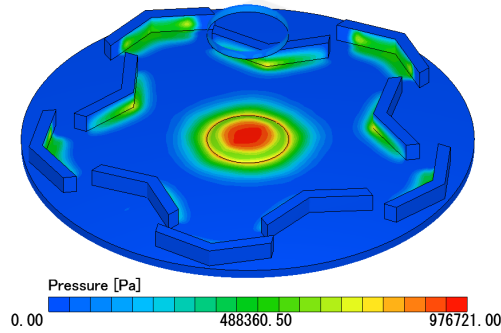


Figure 16: COOL-PAD version 1 pressure plot in scFLOWpost isometric view. Steady-state analysis.

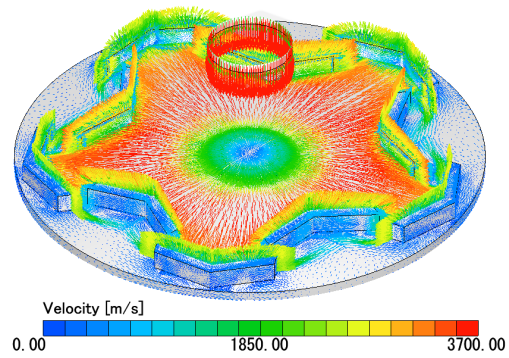


Figure 18: COOL-PAD version 1 flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

Following, in Figure 17, the wall shear stress is presented, with a maximum value of around 8×10^3 Pascals, reaching the highest values on the upper edges of the five V-shape walls looking inwards.

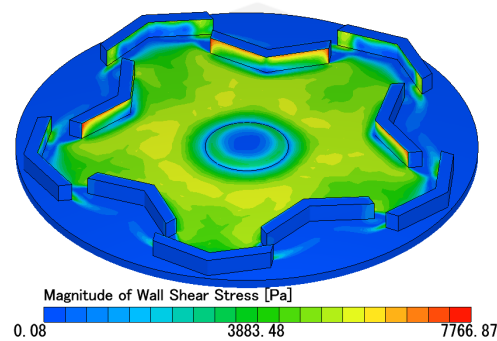


Figure 17: COOL-PAD version 1 wall shear stress plot in scFLOWpost isometric view. Steady-state analysis.

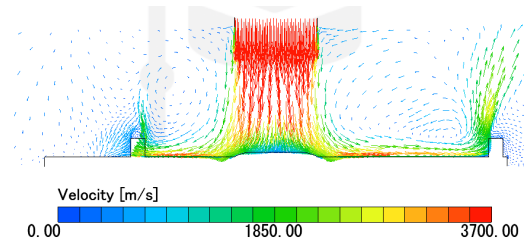


Figure 19: COOL-PAD version 1 velocity plot in scFLOWpost cross-section from Y-axis. Steady-state analysis.

Finally, to conclude with the results of this version 1, the flow velocity is presented in the Figures 18 & 51 below. In these two plots, it can be observed the movement of the flow and its distribution along the platform and, in the cross-section plot, the inclination of the outgoing flow escaping from the Confiner Walls above.

The maximum flow velocity is 3700 m/s that corresponds to the inlet flow condition. However, in order to obtain the velocity of the flow escaping the landing pad, the exiting flow has been calculated at the limit of the Computational Domain (see Figure 43 in Appendix B) that is 75 meters away from the center of the landing pad. This value is around 250 m/s.

2) *COOL-PAD version 2*: This version 2 of the COOL-PAD is an improved design with respect to version 1, as it can be observed in Figure 20. This design has a smoother shape of the Central Hill, this is, the curvature now covers all of the hill surface, removing the flat part from the previous version.

Furthermore, this version has 16 V-shape walls in total, instead of 10 as the previous version, where 8 are radially oriented inwards and the other 8 are oriented outwards. The increment in the number of walls increases the area covered by the walls as well as the amount of vertices where the flow is separated and distributed.

The separation angle between consecutive, same-oriented V-shape walls is 45 degrees and the size of the walls is the same as in the previous version, with the particularity that the corners have been rounded using fillets.

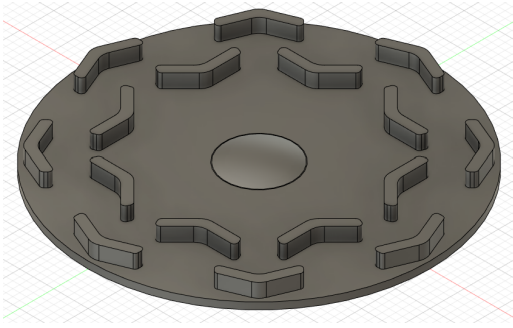


Figure 20: COOL-PAD version 2 3D Model in Fusion 360.

As it is observed in Figure 21, the pressure is also around 1MPa in the Central Hill, rapidly decreasing when moving radially outwards, until it reaches the vertices of the V-shape walls, where it increases considerably, due to the forced redirection of the flow.

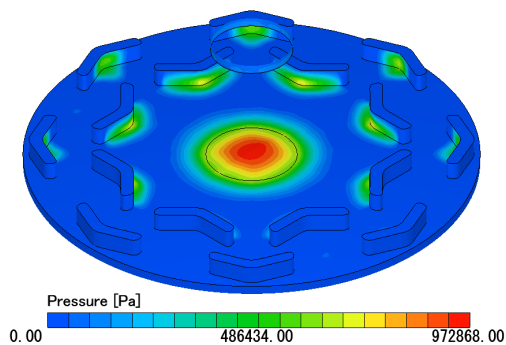


Figure 21: COOL-PAD version 2 pressure plot in scFLOWpost isometric view. Steady-state analysis.

Following, the wall shear stress plot, seen in Figure 22, reveals a high stress zone in between the Central Hill and the inner V-shape walls, however, the highest values of wall shear stress are, like in the first version, located in the lateral edges of the inner V-shape walls. The maximum value is around 6.6×10^3 Pascals.

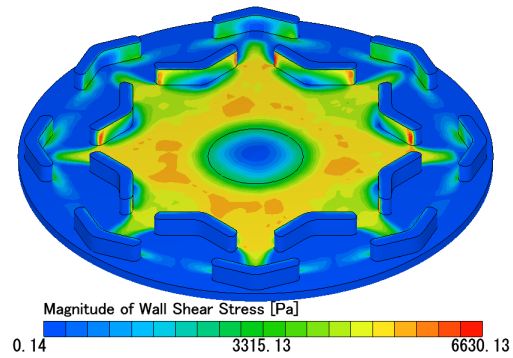


Figure 22: COOL-PAD version 2 wall shear stress plot in scFLOWpost isometric view. Steady-state analysis.

Finally, the flow velocity results show the distribution of the flow along the platform and the cross-sectional plot shows the inclination of the exiting flow from the walls above. The maximum value is at the nozzle (3700 m/s) and the exiting value of the velocity at the Computational Domain limit, 75 meters away from the center, is of 110 m/s (see Figure 44 in Appendix B).

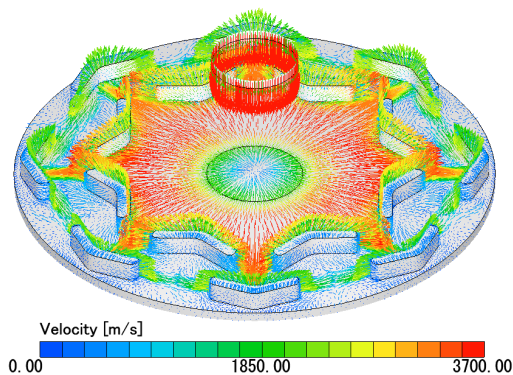


Figure 23: COOL-PAD version 2 flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

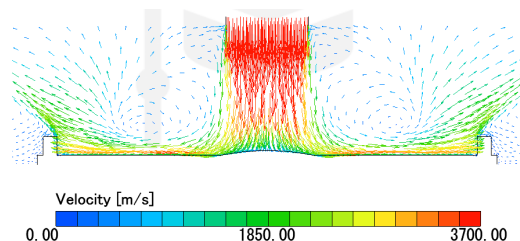


Figure 24: COOL-PAD version 2 velocity plot in scFLOWpost cross-section from Y-axis. Steady-state analysis.

3) *COOL-PAD version 3*: The third version of the COOL-PAD is similar to the version 2 but different types of walls were included additionally, the R-shape walls (rounded) and the I-shape walls. The R-shape walls have been added between the inner V-shape walls in an attempt to address the incoming flow that does not impact in the V-shape walls. Similarly, the I-shape walls are located between the outer V-shape walls in order to further stop the flow from exiting the platform in low angles with respect to the horizontal plane. There are, in total, 8 R-shape walls and 8 I-shape walls.

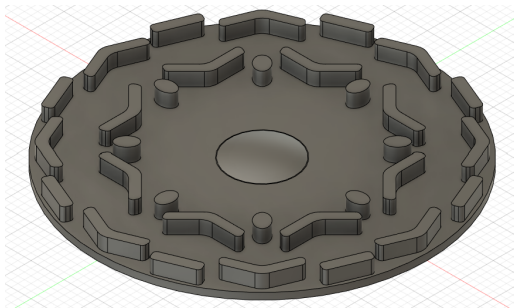


Figure 25: COOL-PAD version 3 3D Model in Fusion 360.

In the pressure plot (Figure 26), it can be observed that the middle part of the Central Hill undergoes the highest pressure, around 1 MPa, directly under the plume. As the flow expands radially, the pressure radically decreases until it reaches the walls. From the plot, it can be observed that the R-shape walls undergo a higher pressure than the V-shape walls.

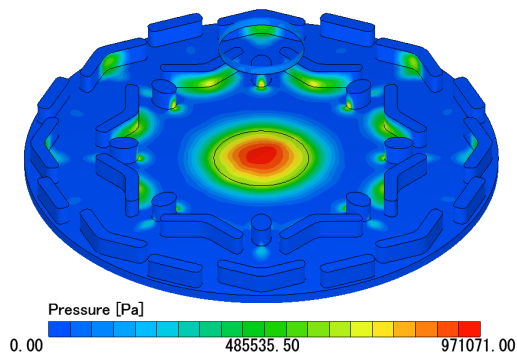


Figure 26: COOL-PAD version 3 pressure plot in scFLOWpost isometric view. Steady-state analysis.

Regarding the wall shear stress plot, the shear stress looks very uniform in between the Central Hill and the R-shape and V-shape walls, however, the highest wall shear stress value (1.067×10^4

Pa) is found in the lower part of the sides of the R-shape walls.

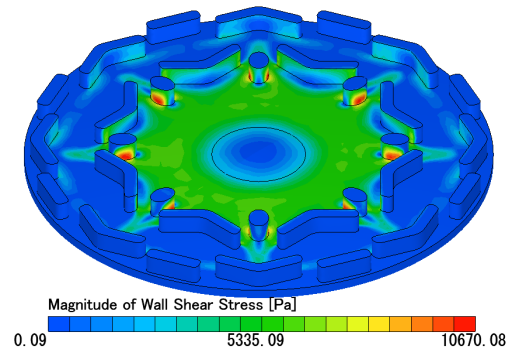


Figure 27: COOL-PAD version 3 wall shear stress plot in scFLOWpost isometric view. Steady-state analysis.

Then, the velocity plots are shown in the figures below, where the flow distribution is shown and it can be observed that there is a boundary in the R-shape walls where the flow velocity is still high, in correlation to the high shear stress in that region. The maximum velocity is at the Nozzle (3700 m/s) and, obtained from Figure 45 in Appendix B, the exit flow velocity at the domain limit is 100 m/s.

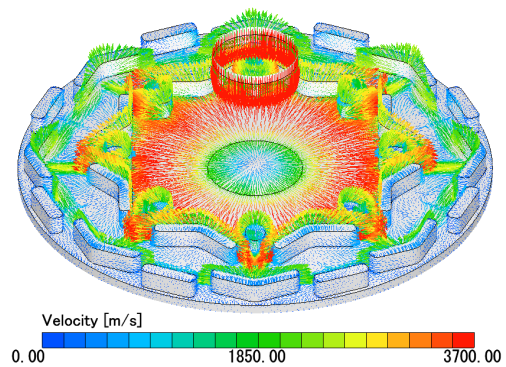


Figure 28: COOL-PAD version 3 flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

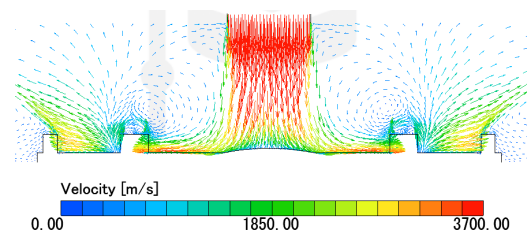


Figure 29: COOL-PAD version 3 velocity plot in scFLOWpost cross-section from Y-axis. Steady-state analysis.

4) *COOL-PAD version 4*: This version of the COOL-PAD is completely different to what has been shown previously, but maintains the philosophy of redirecting the flow. This flower design (shown in Figure 30) aims to redistribute the flow by means of curved walls and bifurcations, in order to reduce its velocity as much as possible. This design has been created with the aid of ellipses, that provided the original curves, which were translated and rotated to create this flower pattern.

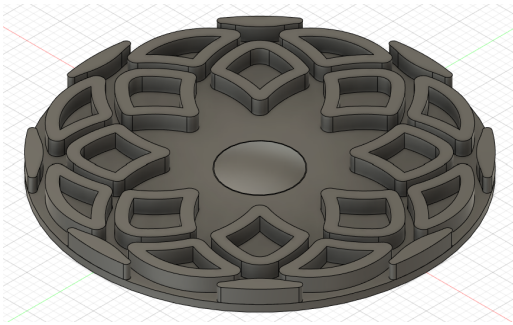


Figure 30: COOL-PAD version 4 3D Model in Fusion 360.

As it is observed in Figure 31, the pressure in the Central Hill is around 1 MPa, whereas it decreases with the radial expansion. The vertices from the first and second layer of walls undergo a high pressure due to the direct opposition to the radial expansion of the flow.

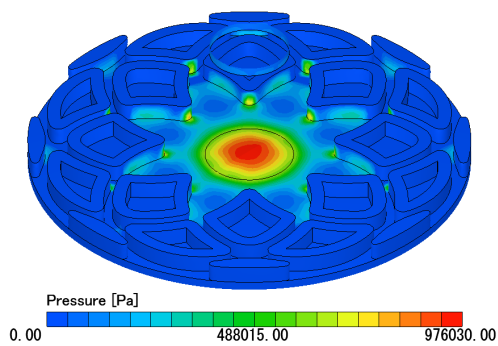


Figure 31: COOL-PAD version 4 pressure plot in scFLOWpost isometric view. Steady-state analysis.

Furthermore, the wall shear stress plot is presented in Figure 32, where it is obtained that the maximum wall shear stress is at the end of the first layer of walls and has a value of 5.7×10^3 Pa. The no man's land region also experiences high wall shear stress, whereas, after the first wall layer, it is drastically reduced.

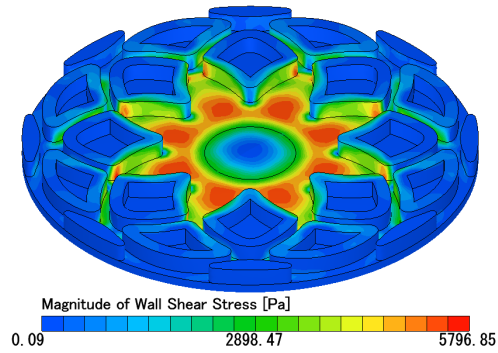


Figure 32: COOL-PAD version 4 wall shear stress plot in scFLOWpost isometric view. Steady-state analysis.

Finally, the flow velocity plots below show the flow distribution along the COOL-PAD, having a high velocity peak in the transition from the first to the second layer of walls. The maximum velocity is at the nozzle (3700 m/s) and the exit flow velocity at the Computational Domain (75 meters from the center of the platform) is of 250 m/s, value obtained from Figure 46 in Appendix B.

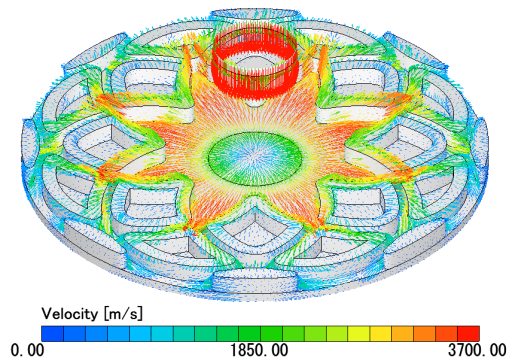


Figure 33: COOL-PAD version 4 flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

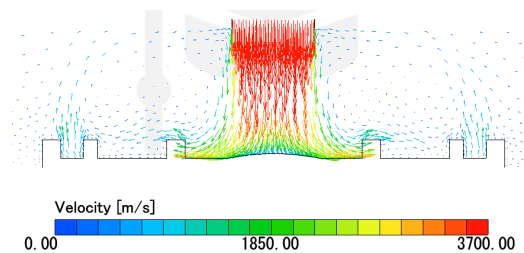


Figure 34: COOL-PAD version 4 velocity plot in scFLOWpost cross-section from Y-axis. Steady-state analysis.

5) *COOL-PAD version 5*: The last version of the COOL-PAD proposed in this project took inspiration from the design shown in Figure 6. This design attempts to redirect the flow and confine it by means of R-shape walls, V-shape walls and C-shape walls, having 8 of each type.

The V-shape and R-shape walls attack directly the flow coming from the Central Hill and guide it through an aperture into a C-shape wall that, at the same time, using its curvature, redirects the flow into the back part of the V-shape wall.

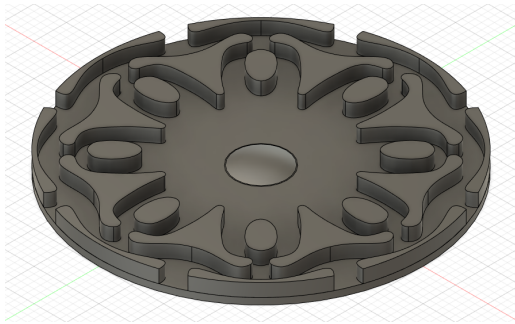


Figure 35: COOL-PAD version 5 3D Model in Fusion 360.

The first result is, as always, the pressure plot (Figure 36), where it is observed that the Central Hill undergoes a pressure of around 1 MPa and it decreases with radial expansion. The R-shape and V-shape walls that attack directly the flow undergo a high pressure in the region oriented to the middle, being the V-shape tip especially high in comparison to the R-shape pressure.

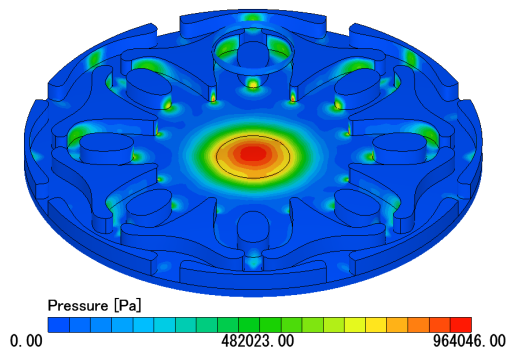


Figure 36: COOL-PAD version 5 pressure plot in scFLOWpost isometric view. Steady-state analysis.

The wall shear stress plot shows a homogeneous distribution along the no man's land, reaching the highest values on the sides of the V-shape walls tip and at the end of the V-shape wall, behind the R-shape wall, whereas the R-shape wall does not

undergo a specifically high shear stress. The maximum value of the wall shear stress is 9.8×10^3 Pa and is located in the V-shape walls.

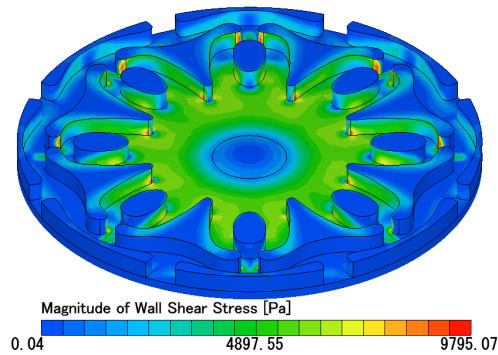


Figure 37: COOL-PAD version 5 wall shear stress plot in scFLOWpost isometric view. Steady-state analysis.

To conclude with this subsection, the Figures 38 & 67 below show the flow velocity distribution along the COOL-PAD version 5. Where it is observed high values on the surroundings of the V-shape walls tips and in the region between the R-shape, C-shape and end of V-shape walls. The highest velocity is at the Nozzle (3700 m/s) and the exit velocity at the Computational Domain (obtained from Figure 47 in Appendix B) has a value of 217 m/s.

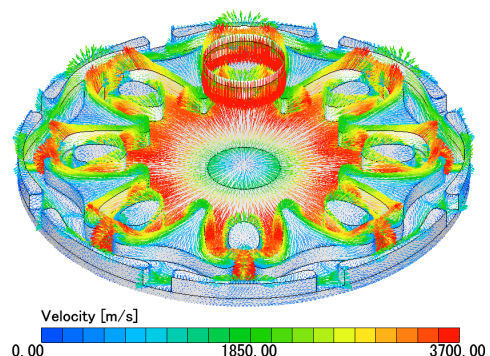


Figure 38: COOL-PAD version 5 flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

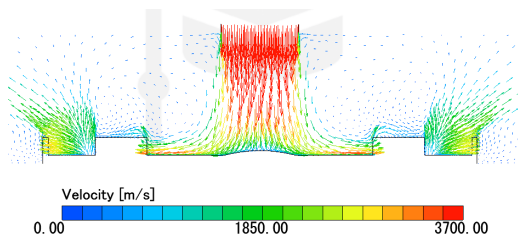


Figure 39: COOL-PAD version 5 velocity plot in scFLOWpost cross-section from Y-axis. Steady-state analysis.

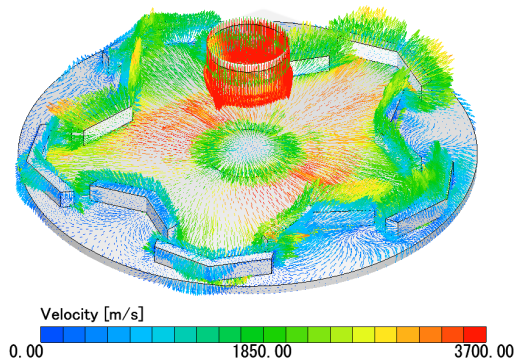


Figure 40: COOL-PAD version 1 velocity plot in scFLOWpost. Transient analysis.

B. Results: Transient Analysis

The transient analysis has been performed during 15 seconds in 200 cycles, having a delta of 0.075 seconds per cycle. Regarding this transient analysis, some inconsistencies have been encountered that significantly decreases the reliability of these results.

The problematic with the transient results consist on a non-homogeneous spread of the flux, pressure and wall shear stress. For further analysis, the cycles in between were also analyzed with the aim to find a guideline that could explain the non-homogeneous results obtained in the last cycle. This way, the solver was asked to provide results every 10 cycles of the computation (every 0.75 seconds) and, after that, those results were put together in a visual sequence with the postprocessor scFLOWpost.

Obviously, the non-homogeneity of the results is very noticeable, because the 3D models of the different versions of COOL-PAD have rotational symmetry, hence it does not make sense to have areas that undergo much more pressure or shear stress than similar areas rotated by a certain angle. In addition, the partial results of the cycles in between does not follow a coherent guideline, some looks homogeneous while others do not.

To put an example, the following figures will show these inconsistencies described above:

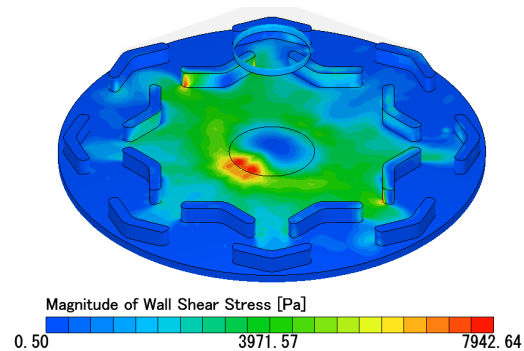


Figure 41: COOL-PAD version 2 wall shear stress plot in scFLOWpost. Transient analysis.

Here, the scalar differences along the surface of the platform can be easily observed. The rest of the results of the transient analysis are exposed in Appendix C.

There are two main identified reasons why the transient analysis is not reliable:

- **Mesh:** The mesh generated for this type of analysis is not small and refined enough, therefore, it generates some inconsistencies during the computation. The reason why this happened is due to the limitation in computational resources, where the mesh of the model was enlarged in order to reduce the big computational time needed for the calculation.
- **Cycles:** The time step is too large for this analysis, giving a small number of cycles (200 in this case). This number of cycles has to be enlarged, but, again, because of the limited computational resources, this was not possible in order to maintain reasonable computational times.

The time step is closely related to the Courant Number. Courant number is a parameter used for determining the time step and is defined by Equation 12 [41]:

$$C = \frac{u\Delta t}{\Delta L} \quad (12)$$

Where u is the flow velocity, Δt is the time step and ΔL is the increment in longitude, in this case it corresponds to the mesh size. For a good analysis it is a matter of interest to maintain the Courant Number as small as possible. Since the flow velocity is very high (3700 m/s) and the time step was very large, it is easy to deduce that the Courant Number of this analysis is high.

Thus, the only way to address this problem is to reduce the time step. The mesh size and time step influences directly into this number and, because of the mesh varies in size depending on the area and, additionally, it must faithfully represent the geometry of the model, the importance relies on the time step, that must be reduced as much as possible, but, due to the reasons explained before, it was not feasible to carry out.

C. Discussion

During this research, some characteristics for the design of lunar landing pads have been tested and, with the results of these analyses, many hypothesis are proposed for results interpretation.

First of all, regarding the pressure results, the Central Hill performs better when it is fully rounded in the middle and with a smoother transition to the flat part of the platform. It can be observed that there is a small orange ring of higher pressure around the hill in COOL-PAD v1 (Figure 16), whereas the other versions does not exhibit it. Remember that the hill in version 1 is different to the hills from the versions 2 to 4.

Another interesting observation with respect to the pressure plots are the surfaces opposing directly to the horizontal flow, after it is redirected by the hill. The walls with wider surfaces facing the flow undergo less pressure concentration than the ones that exhibit sharper surfaces, as it is observed in the R-shape walls in Figure 26, the flower walls in Figure 31 and the tip of the V-shape and R-shape walls in Figure 36 (corresponding to versions 3, 4 & 5, respectively). Therefore, the wider the surface facing the flow, the less pressure concentration, but, on the other hand, the higher force acting on the whole wall. Finally, the

pressure along the flat part of version 4, is observed to be higher than in the other designs, because the Confiner Walls are closer to the center of the platform, attacking the flow earlier and reducing the free space of that region.

Secondly, the wall shear stress plots seem to show that the regions where the pressure is high, the wall shear stress is not. The magnitude of this variable is highly correlated to the magnitude of the flow velocity along the landing pad.

Comparing the versions 1 and 2, the value of the wall shear stress at the inner V-shape walls is lower in version 2, indicating that rounding the walls and generating curved edges provides a better performance against wall shear, reducing its magnitude.

Furthermore, it is observed that the sharp shapes in versions 3, 4 & 5 undergo a high shear stress relative to its overall level. The peak of this is observed in version 3 (Figure 27) where the R-shape walls undergo a very high shear stress compared to the other designs. This is because the flow has free space along the center of the platform and arrives to that point with a high velocity, moreover, the R-shape wall is small enough so the flow is able to easily go around it at high speed.

The designs that perform better are the ones that have more surface area attacking the flow and less directly exposed area oriented with small angles with respect to the flow radial expansion.

Regarding the flow velocity along the landing pads there are very interesting results. As it is observed in the versions 1, 2 & 3, the flow that remains in the platform is, first, slowed and redirected by the inner walls and, secondly, stopped by the outer walls. On the other hand, version 4 is only redirecting the flow, without obstructing it, therefore, in Figure 33 it can be observed that the flow exiting the platform moves at a certain speed, in contrast to the other plots.

The exit flow values obtained from the Computational Domain velocity plots are representative from what is described above: The velocities decreases from versions 1 to 3, whereas the exit flow velocity in version 4 is as high as version 1, which is a much simpler design. With respect to COOL-PAD v5 (Figure 38) the exit flow velocity at the limit of the Computational Domain is also high, but smaller than versions 1 & 4. This could be caused due to the late obstruction of the flow. As it is observed in the

plot, the flow reaches the end of the V-shape wall with high speed, producing its escape above the walls and reaching the outer C-shape walls with relatively high velocity.

With these analyses, it is proved that the COOL-PAD prevents the formation of orbiting clouds and mitigates sandblasting of surrounding structures, even if some of the inner parts of the landing pad are eroded and introduced in the flow.

In the Table II below, a summary of the results can be observed. This table shows the values of pressure, wall shear stress, exit flow velocity and volume of every COOL-PAD version presented in this document.

Version	P [MPa]	τ [Pa]	v [m/s]	V [m ³]
1	0.977	7766.9	251.3	2376.5
2	0.973	6630.1	111.2	2416.8
3	0.971	10670.1	100.5	2637.3
4	0.976	5796.9	249.7	3457.9
5	0.964	9795.1	217.1	3249.1

Table II: COOL-PAD results version comparison.

Looking at the table above, it is concluded that the version which has the best overall performance is COOL-PAD v2 (Figure 20) because it has the second lowest shear stress, exit flow velocity and volume, becoming the best balanced version among the five. In addition, the velocity, which is very important to avoid sandblasting, is very close to the lowest (10 m/s away), as well as its volume is very low in comparison to the versions 3, 4 & 5. Therefore, the winner among these contenders is COOL-PAD v2.

Finally, as a last result iteration, the shear strength of the regolith has been computed using Equation 1 from Section II-C and the following optimistic data [42]:

- Cohesion (c): 1000 Pa.
- Internal Friction Angle (φ): 50 degrees.
- Pressure Normal (σ): 1000 Pa.

The shear strength τ of the regolith obtained with this data is around 2190 Pa. The following figure (Figure 69, and just for additional visual information, shows the COOL-PAD version 2 shear stress plot, but, as it is observed, the colored areas are the zones where the shear stress exceeds the shear strength of the lunar regolith, meaning that these are the risk areas to be reinforced by sintering stronger materials. The rest of these plots are exposed in Appendix D.

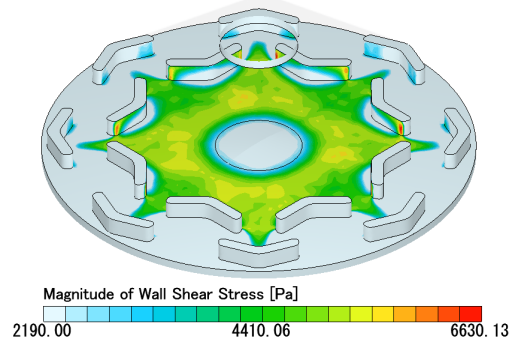


Figure 42: COOL-PAD version 2 wall shear stress plot against lunar regolith shear strength in scFLOWpost. Steady-state analysis.

V. CONCLUSIONS

In this final section, the conclusions of the work explained in this document are going to be exposed.

- The exit velocity values presented in Table II are way lower than the lunar escape velocity (2.4 km/s), hence, the designs proposed here prevent the formation of orbiting dust clouds and mitigate sandblasting effects.
- When the flow velocity is high, sharp shapes and surfaces oriented at small angles with respect to the radial component undergo specially high shear stresses. Better performance is obtained with wide surfaces and rounded edges.
- Direct opposition to the flow avoiding abrupt shape changes in high velocity areas provides better performance.
- The best performance found in this research has wide convex surfaces in the inner part of the platform and wide concave surfaces in the outer part. The wide convex surfaces obstruct and redirect the flow to the concave surfaces, confining the flow avoiding unnecessary stress peaks.
- Rounded Central Hill is important for distributing the pressure and flow horizontally along the platform.
- Transient analysis was not reliable and it requires further computational resources.

VI. FUTURE WORK

The future work to be done after this project is:

- Iteration and optimization of the designs and analyses for further correlation of results.
- Carry out the transient analysis with reliable results.
- Perform more detailed analyses taking into account temperatures, heat transfer, mixed gas, etc.

- Carry out analyses at different heights (5, 10, 20 meters for instance) and during various times.
- Perform a mechanical and thermal analysis in order to check the structural stability.
- Study of the viability for construction with in-situ resources.

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APPENDIX

A.

BUND command lines included in the SPH file to improve the element stabilization tolerance:

```
#
BUND
  VELX 0 1e20 -1e20 3 1e40 -1e40
  VELY 0 1e20 -1e20 3 1e40 -1e40
  VELZ 0 1e20 -1e20 3 1e40 -1e40
  PRES 0 1e20 -1e20 3 1e40 -1e40
  TEMP 0 1e20 -1e20 3 1e40 -1e40
  TURK 0 1e20 -1e20 3 1e40 -1e40
  TEPS 0 1e20 -1e20 3 1e40 -1e40
/
```

B.

Computational Domain exit flow velocity figures used to obtain the flow velocity exiting the platform of the COOL-PAD (Steady-state analysis):

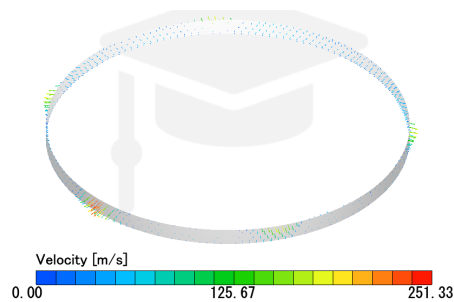


Figure 43: COOL-PAD version 1 Computational Domain exit flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

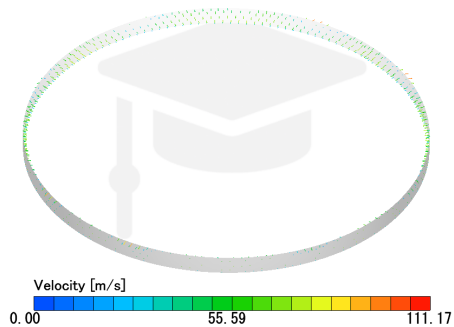


Figure 44: COOL-PAD version 2 Computational Domain exit flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

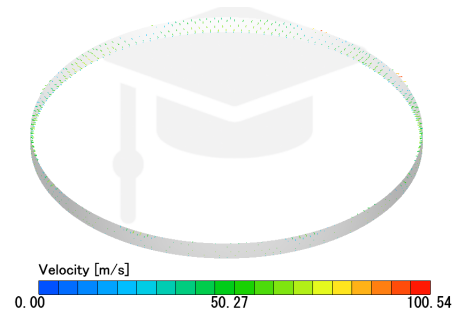


Figure 45: COOL-PAD version 3 Computational Domain exit flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

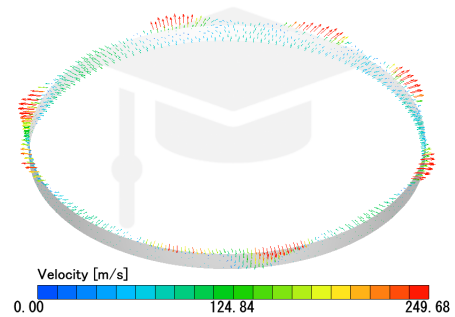


Figure 46: COOL-PAD version 4 Computational Domain exit flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

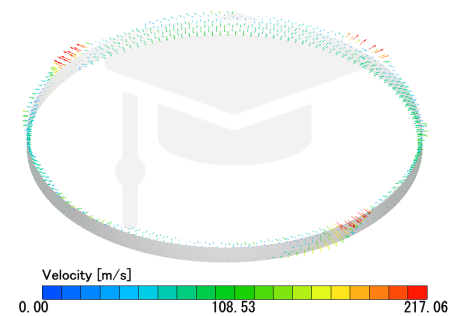


Figure 47: COOL-PAD version 5 Computational Domain exit flow velocity plot in scFLOWpost isometric view. Steady-state analysis.

C.

Pressure, wall shear stress and velocity figures obtained from the COOL-PAD transient analysis:

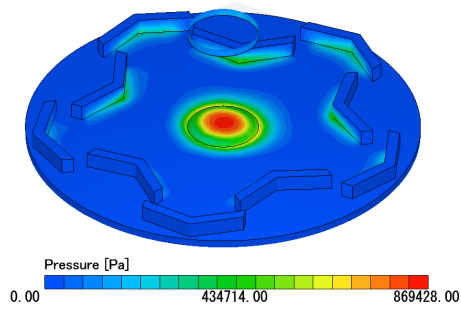


Figure 48: COOL-PAD version 1 pressure plot in scFLOWpost isometric view. Transient analysis.

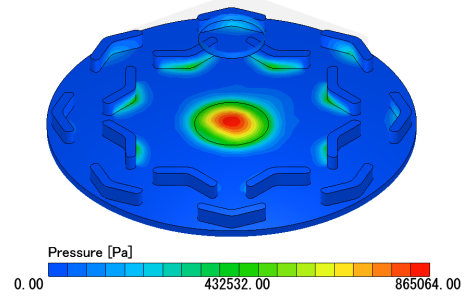


Figure 52: COOL-PAD version 2 pressure plot in scFLOWpost isometric view. Transient analysis.

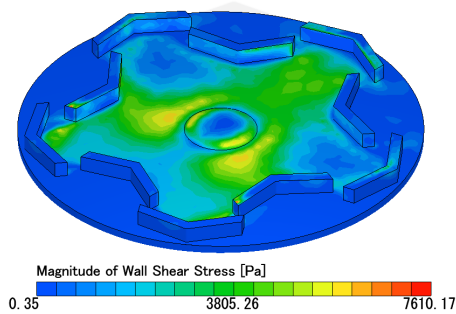


Figure 49: COOL-PAD version 1 wall shear stress plot in scFLOWpost isometric view. Transient analysis.

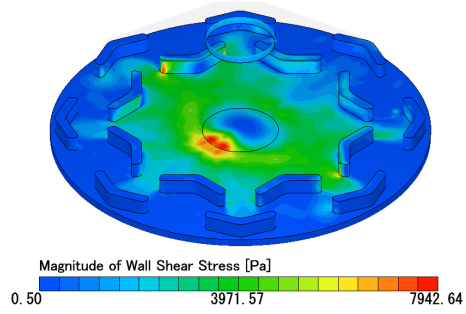


Figure 53: COOL-PAD version 2 wall shear stress plot in scFLOWpost isometric view. Transient analysis.

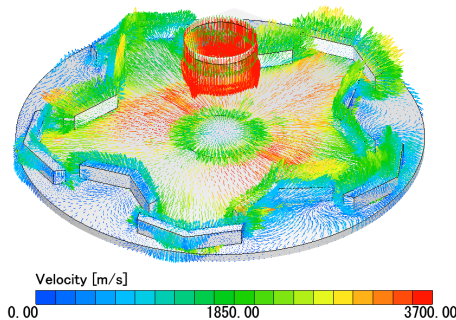


Figure 50: COOL-PAD version 1 flow velocity plot in scFLOWpost isometric view. Transient analysis.

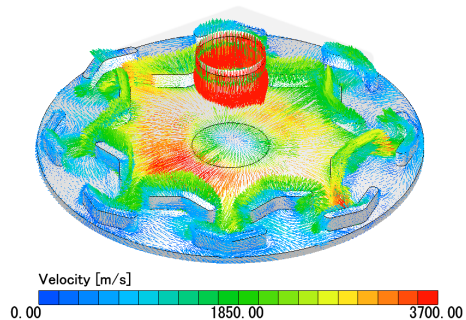


Figure 54: COOL-PAD version 2 flow velocity plot in scFLOWpost isometric view. Transient analysis.

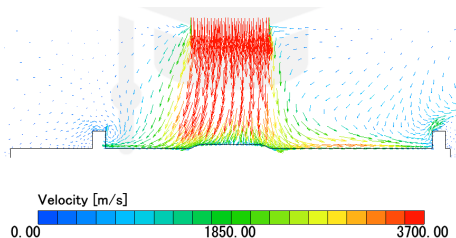


Figure 51: COOL-PAD version 1 velocity plot in scFLOWpost cross-section from Y-axis. Transient analysis.

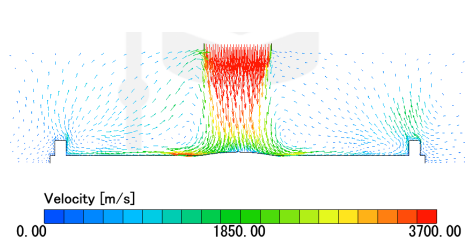


Figure 55: COOL-PAD version 2 velocity plot in scFLOWpost cross-section from Y-axis. Transient analysis.

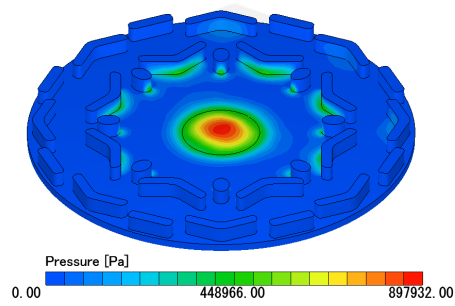


Figure 56: COOL-PAD version 3 pressure plot in scFLOWpost isometric view. Transient analysis.

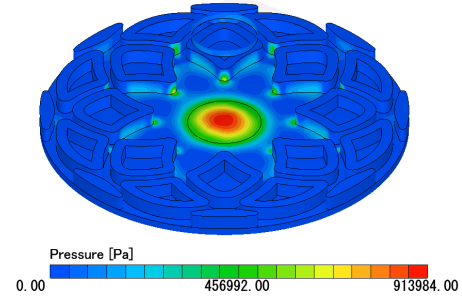


Figure 60: COOL-PAD version 4 pressure plot in scFLOWpost isometric view. Transient analysis.

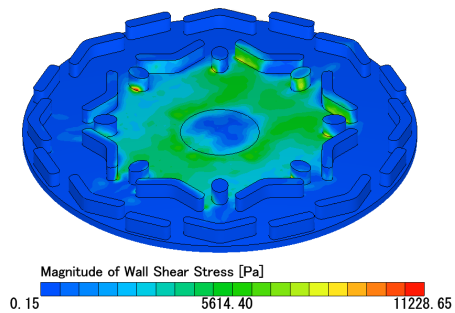


Figure 57: COOL-PAD version 3 wall shear stress plot in scFLOWpost isometric view. Transient analysis.

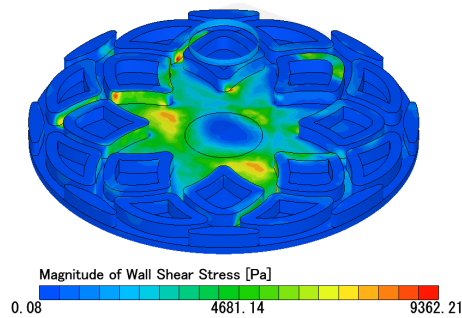


Figure 61: COOL-PAD version 4 wall shear stress plot in scFLOWpost isometric view. Transient analysis.

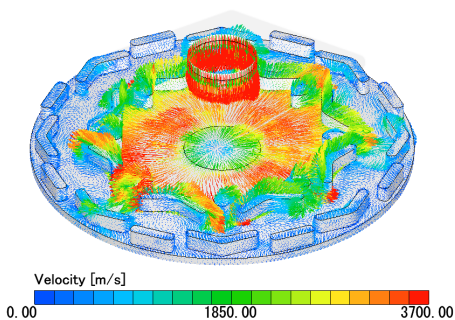


Figure 58: COOL-PAD version 3 flow velocity plot in scFLOWpost isometric view. Transient analysis.

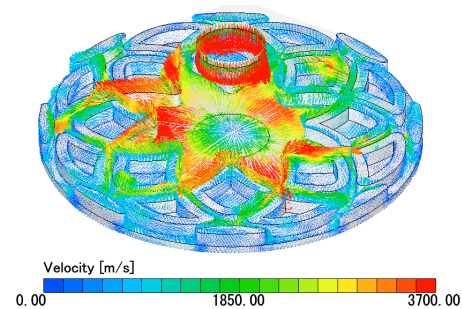


Figure 62: COOL-PAD version 4 flow velocity plot in scFLOWpost isometric view. Transient analysis.

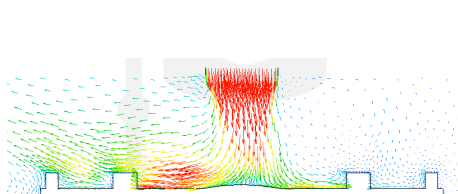


Figure 59: COOL-PAD version 3 velocity plot in scFLOWpost cross-section from Y-axis. Transient analysis.

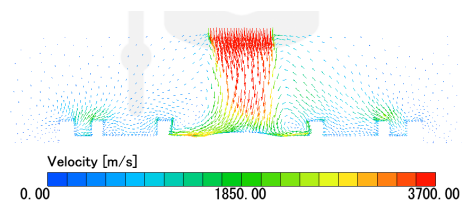


Figure 63: COOL-PAD version 4 velocity plot in scFLOWpost cross-section from Y-axis. Transient analysis.

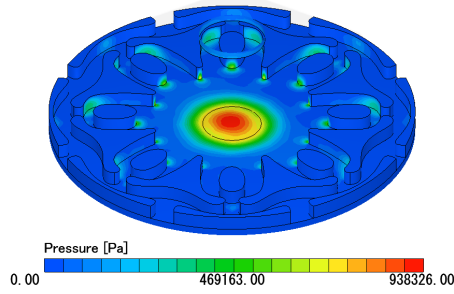


Figure 64: COOL-PAD version 5 pressure plot in scFLOWpost isometric view. Transient analysis.

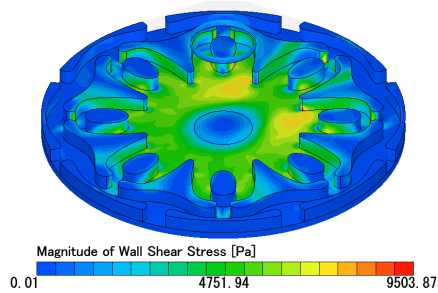


Figure 65: COOL-PAD version 5 wall shear stress plot in scFLOWpost isometric view. Transient analysis.

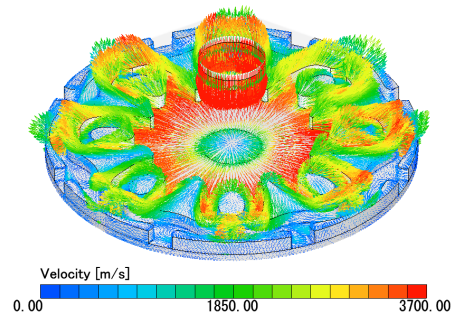


Figure 66: COOL-PAD version 5 flow velocity plot in scFLOWpost isometric view. Transient analysis.

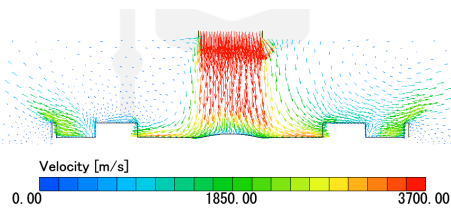


Figure 67: COOL-PAD version 5 velocity plot in scFLOWpost cross-section from Y-axis. Transient analysis.

D.

Wall shear stress against regolith shear strength plots for all versions of the COOL-PAD

(Steady-state analysis):

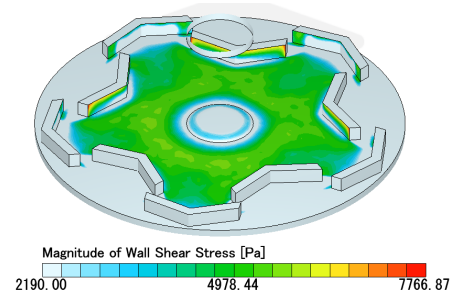


Figure 68: COOL-PAD version 1 wall shear stress plot against lunar regolith shear strength in scFLOWpost. Steady-state analysis.

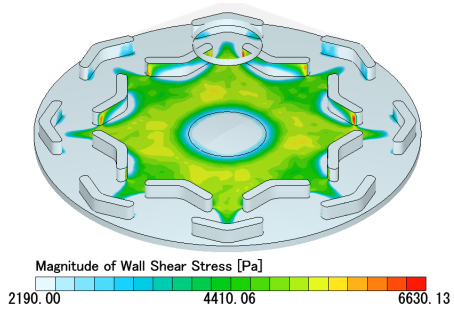


Figure 69: COOL-PAD version 2 wall shear stress plot against lunar regolith shear strength in scFLOWpost. Steady-state analysis.

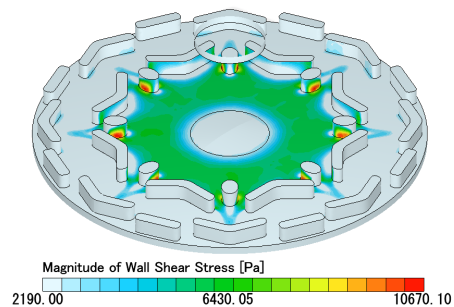


Figure 70: COOL-PAD version 3 wall shear stress plot against lunar regolith shear strength in scFLOWpost. Steady-state analysis.

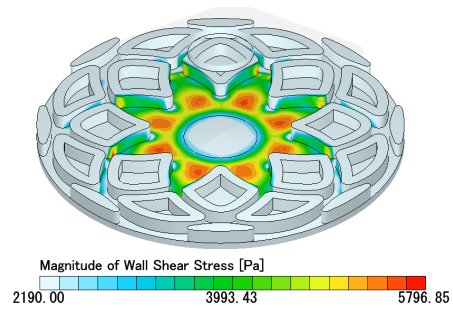


Figure 71: COOL-PAD version 4 wall shear stress plot against lunar regolith shear strength in scFLOWpost. Steady-state analysis.

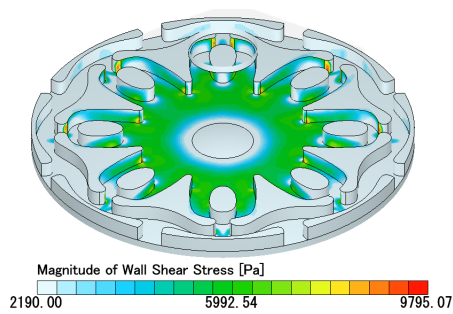


Figure 72: COOL-PAD version 5 wall shear stress plot against lunar regolith shear strength in scFLOWpost. Steady-state analysis.

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